NEXIAM: A ROLE-BASED COLLEGE ADMISSION TESTING SYSTEM FOR THE GUIDANCE AND REGISTRAR OFFICES AT ISPSC TAGUDIN

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Chapter 1

INTRODUCTION

Background of the Study

Assessment serves two purposes. The first is validation which gives feedback to students on what they have learned. The second is academic filtering that tests whether learners can excel in a certain field, or in this case, whether they are aligned with the university course they intend to pursue. Assessments remain prevalent in higher education. A 2024 scoping review by French et al. found that although assessments offer pedagogical benefits like stronger knowledge retention and increased motivation for some learners, these benefits are often outweighed by drawbacks such as heightened student anxiety, surface-level learning, and equity concerns. This means assessment design continues to receive scrutiny, including the psychometric properties of examinations. However, besides psychometric issues, there is also a gap concerning the operational logistics of administering examinations at scale. This operational perspective is comparatively under-researched, especially from the standpoint of institutional impact (Woelert, 2023).

This revelation along with the researchers' background sets the direction of this study to solve the bottlenecks on the non-psychometric side of admissions, specifically the infrastructure surrounding the admission process. Patil et al. (2021) found that large-scale examinations and admissions encounter challenges with record integrity, applicant verification, and resource coordination, particularly when paper-based workflows are used. Sánchez-Mendiola et al. (2025) demonstrated such challenge when analyzing results from over 600,000 examinees on one large-scale

exam. In contrast, Thakur et al. (2025), through their TrueGradeAI framework, showed that digital admission pipelines do more than automate scoring. They introduce architectural patterns that reduce grading bias and human subjectivity. The point is that while psychometric properties get considerable attention, the operational side of admissions remains under-invested despite being the component responsible for administering examinations. As a result, institutions often lack the downstream structures needed to process scores and deliver results efficiently.

Liu et al. (2025) further explained that conventional admissions still rely on fixed historical yield assumptions while overlooking applicant behavior uncertainty. This means prolonged decision timelines distort enrollment projections and make it harder for institutions to convert admissions into enrollment. Esquivel and Esquivel (2020) demonstrated how predictive modeling is used to estimate class sizes from an already admitted but unenrolled students in a Philippine university. This weighs heavy for an institution's resource allocation and projections. After results are released, speed matters from the applicant's perspective because many applicants receive multiple offers and will commit to whichever institution processes them faster (Ruffalo Noel Levitz, 2023). Slow admissions are not merely an inconvenience. They are structural barriers. For a public state college on a fixed enrollment calendar, every day spent through inefficient scoring and various delays, is a day where potential applicants could consider another institution, especially in the perspective and bias of efficient service delivery. This necessitates a pipeline that is fast on top of an effective delivery.

In the global landscape, admission systems have also moved toward innovation.

Ever since the COVID-19 pandemic, many institutions have fast tracked their adoption of artificial intelligence and cloud platforms to modernize administrative services (EDUCAUSE, 2023; Marinoni et al., 2020). Likewise, the application of machine learning to admission processes is also becoming more relevant. Lira et al. (2023), publishing in Science Advances, found that machine learning models can evaluate applicants using unstructured application materials like essays, and that these assessments correlate with long-term life outcomes. This strengthens the case for AI within admissions. However, Van Busum and Fang (2025) examined bias in AI admission models and raised fairness concerns about algorithm-assisted decision-making. Therefore, AI should not completely replace human judgment but instead function as part of a hybrid model where professional judgment remains central.

With regards to unified admission platforms, Adwan et al. (2023) developed a centralized mobile-based admission application for universities in the GCC region. Their system achieved a 97.7% usability using Nielsen heuristics, which shows that unified admission platforms are practical across institutions. Inferring from this, a unified admission pipeline combines several operational features into one platform to address such admission-related bottlenecks. One onset feature is AI-powered chatbots, which is shown to handle routine applicant questions consistently at scale (Labadze et al., 2023). Another feature is role-based access control, which is important for keeping educational data secure at an institutional scale (Sandhu et al., 1996; Kabier et al., 2023).

Nationally, the Philippines' digitalization has been driven more by policy rather than by market pressure. The CHED Strategic Plan (CHED, 2022) sets such

institutional direction. Rooting from the Republic Act No. 10931 (2017) which increased admission volumes after removing tuition fees in public higher education, putting more pressure on staff and infrastructure. Alongside Republic Act No. 10844 (2016) which established the DICT, the pressure makes it more real as it makes ICT policy a government-wide mandate. In hindsight, the Republic Act No. 10173 (2012) adds to this pressure the requirement to secure handling of applicant records, which paper systems cannot fully support. These laws make digitization both necessary and challenging. The National Privacy Commission (2020) has issued advisories for the education sector on handling applicant records. On top of this, Villarino (2025) found that students in provincial state colleges already use consumer AI tools ahead of their own institutions. Ilagan et al. (2025) developed a virtual support agent for Filipino university students, showing that LLM-based student services are deployable in the Philippine HEI context. This creates a governance gap that bypasses proper compliance with Republic Act No. 10173, despite students being ready and AI-literate. Admissions should therefore take advantage of this familiarity by using students' AI literacy into institutional applications instead of leaving it under-utilized. AI chatbots are one application of how this can work.

Another relevant point is that every feature within a unified admission pipeline should first be validated independently. Cuerdo et al. (2021) validated EvalBee, an optical mark recognition tool, in a Philippine public school. Earlier, Catalan (2017) demonstrated automated multiple-choice scoring using readily available software at De La Salle University, tested on 800 answer sheets across eight courses. Dayata and Yap (2024) deployed a retrieval-augmented generation pipeline at the University of

San Carlos, published at IEEE COMNETSAT. Tan (2025) validated data-driven optimization through K-Means algorithm for student course selection at an ACM conference. Mittal and Malhotra (2026) confirmed through a comprehensive survey that clustering-based recommendation approaches remain a staple in this domain. Together, these independent solutions support the development of an integrated admission pipeline.

Locally, ISPSC Tagudin reflects these national issues. The campus processes about 700 to 1,000 freshman applicants every year using a fully manual workflow covering intake, examination, and result delivery. Previous campus studies have tried to improve institutional processes, but none addressed admissions directly. Yukee et al. (2025) validated K-Means profiling on the campus's own entrance exam data. Ballesteros et al. (2025) confirmed that K-Means works for student profiling at ISPSC. Eustaquio and Nisperos (2023) built a web-based registration system with decision-support for billing, grading, scholarship, and retention tracking.

These studies show that data-driven decision-support systems already exist on campus. Eustaquio and Nisperos in particular demonstrate that a single digital platform can support multiple administrative functions, providing a conceptual basis for an integrated admission platform. However, their system focused on post-admission processes, not the admission testing phase itself. Angala et al. (2023) implemented a document management system, and Cenezan and Gacusan (2025) built analytics-driven records management.

The campus does not lack successful digitization projects or local research. What it lacks is a unified admission testing system, one platform that can scale and be

architecturally integrated, bringing together scoring, aptitude assessment, applicant classification, and structured guidance within one pipeline. As presented in various studies, these pieces are scattered. Yukee's clustering work is deployed but lacks features for admission pipeline inclusion. Cuerdo's OMR validation sits in a journal. Dayata's chatbot is proven to work and is ready for replication. All these features, altogether, have never been connected into one pipeline that is proven to work on an admission process and in the context of how Guidance and Registrar can manage these during an actual admission season. Dash et al. (2026) proposed a similar AI-assisted unified platform for college applications but stopped at the design stage, confirming that the gap is in deployment and governance, not in concept. This integration has to comply with Republic Act No. 10173 by securing both physical records and digitally processed applicant information, especially when data is processed externally (Slade and Prinsloo, 2013). For the AI component, retrieval-augmented generation should draw from institutional policy documents rather than unrestricted language models to reduce hallucinations (Lewis et al., 2020).

Therefore, a unified, role-governed admission testing system is the gap that surfaces from these literature reviews and contemporary necessities. A platform that can integrate applicant registration, streamlined testing operations, automated scoring, decision-support with AI advisory services, and result delivery into one admission ecosystem serving the Guidance Office, Registrar, and other institutional stakeholders. Finally, this needs-finding surfaces two fundamental questions: whether a unified platform outperforms existing traditional processes, and whether it is acceptable according to its intended purpose across different users and use cases.

Conceptual Framework of the Study

NEXIAM uses an IPO model as its framework as this fits the descriptive-developmental study. The input serves as variables that covers campus data like applicant profiles, exam scores, and operational workflow data. Following that is the process which iterates on the development through the defined software model, and detailed through the project planning and assignments. After the development stage, the system will be deployed and tested. This phase will be the conduct of Technology Acceptance Model or TAM. This will measure NEXIAM against the current process and while being part of the process stage, TAM will transform system use to measurable evidence. Lastly, the output variables will be traced through two connected parts. First is the admission testing system platform itself and the second one is the research results from the paired TAM measures before and after deployment. These results will prove whether NEXIAM outperforms traditional methods with statistical backing. This conceptual framework combines system building and user testing into one design, fulfilling the objectives as a whole.

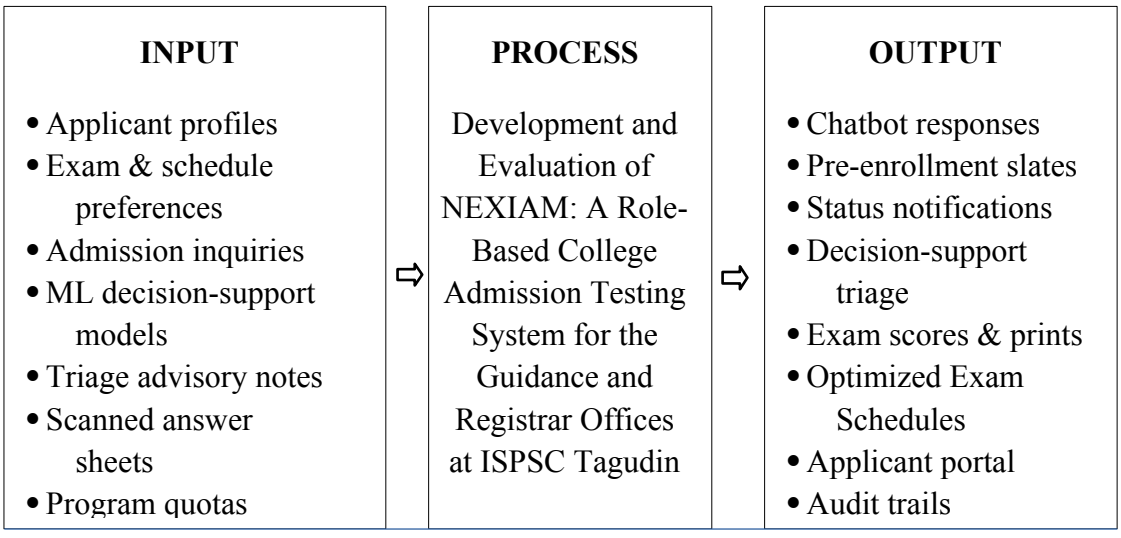


Figure 1: Input-Process-Output Diagram

Objectives of the Study

This study aims to develop NEXIAM: A Role-Based College Admission Testing System for the Guidance and Registrar Offices at ISPSC Tagudin Campus. The web-based platform centralizes the admission testing pipeline that composes of applicant registration, examination, scoring, decision-support and result delivery.

More specifically, this study seeks to accomplish the following:

1. To analyze the existing college admission testing processes, operational workflows, and process gaps at ISPSC Tagudin Campus through descriptive inquiry.
2. To determine the system requirements of NEXIAM in terms of functional requirements and system features needed to address the identified process gaps.
3. To evaluate the acceptance of NEXIAM using the Technology Acceptance Model (TAM) in terms of perceived usefulness, perceived ease of use, attitude toward use, and behavioral intention to use.
4. To determine whether there is a significant difference between the existing manual admission testing process and NEXIAM in terms of perceived usefulness and perceived ease of use.

Null Hypotheses

H_{01}: There is no significant difference in the perceived usefulness between the existing manual admission testing process and NEXIAM.

H_{02}: There is no significant difference in the perceived ease of use between the existing manual admission testing process and NEXIAM.

Test: Paired samples t-test, with the Wilcoxon signed-rank test as a non-parametric alternative contingent on Shapiro-Wilk normality results | α = .05 | Effect size: Cohen's d (Cohen, 1988) (parametric case) or matched-pairs rank-biserial correlation (non-parametric case)

Scope and Limitation of the Study

This research covers the design, development, and evaluation of NEXIAM for managing the college admission testing workflow at the Guidance and Registrar Offices of ISPSC Tagudin Campus, for the admission cycle running June through September 2026. The study covers the full admission testing pipeline from applicant registration through result delivery and post-examination counseling support, across seven user roles governed by Role-Based Access Control (RBAC). A Proctor actor perspective is additionally reflected in the system's use case model, representing a duty overlay assignable to any existing staff role at administrative discretion rather than a distinct eighth role.

The study is strictly bounded to the admission testing function. Enrollment processing, tuition management, long-term academic records, and cross-campus resource allocation are outside scope. System deployment is restricted to an external cloud-hosted environment, a decision driven by local campus hardware constraints. The proctoring interface is designed with offline resilience for examination venues where network connectivity may be unavailable or intermittent, a condition that reflects the documented infrastructure reality of ISPSC Tagudin Campus, where network access is concentrated in designated office areas and examination venues may have limited or no reliable connectivity. The proctoring interface uses service-

worker caching to maintain examination session state during connectivity loss, with synchronization to the cloud database upon connection restoration. All other system functions outside the proctoring interface require cloud connectivity.

AI inference for the hybrid RAG chatbot and the human-gated AI scheduling assistant relies on free-tier external model access, which bounds the volume of concurrent inference requests the system can sustain. Paid or dedicated inference capacity is outside this study's scope.

Methodologically, because the September 2026 evaluation window occurs outside the active admission cycle, purposive sampling utilizes simulated evaluators executing role-based scenarios to simulate live operational conditions, measured via paired pre- and post-test Technology Acceptance Model (TAM) questionnaires. The study's evaluation window therefore cannot produce findings based on live applicant behavior, and results are accordingly bounded by simulated operational conditions.

While NEXIAM is designed to support compliance with Republic Act No. 10173 through Role-Based Access Control and application-layer data governance, formal National Privacy Commission (NPC) certification and the Privacy Impact Assessment required on the machine learning model effectiveness for applicant profiling fall outside this study's scope. Applicant consent for digital processing is assumed to be covered under the existing ICAT application form's data collection procedures, which applicants complete as part of the standard admission process; formal verification of this consent basis is identified as future work alongside the NPC certification and Privacy Impact Assessment noted above.

Importance of the Study

The Community. Local residents benefit from a streamlined admission process that reduces the financial and time costs associated with multiple mandatory campus visits. By digitizing result retrieval and notifications, the system lowers access barriers for applicants living in distant municipalities.

The Registrar Office. Automated intake and scheduling tools replace manual document encoding with a traceable, digital workflow for better inter-office coordination. Role-based data isolation supports the office's compliance posture under Republic Act No. 10173 by operationalizing application-layer data protection aligned with the Act's key obligations.

The Guidance Office. Computer-vision scoring and live data triage automate time-consuming clerical tasks, allowing counselors to focus on professional student placement and profile analysis. The chatbot and digital audit trails further enhance office efficiency and data accountability.

The Program Heads. By shifting quota management to a real-time dashboard, Program Heads are freed from fielding continuous, informal status inquiries from counselors during active consultation sessions, gaining an immediate overview of their department's enrollment capacity.

The Applicants. Applicants gain access to a self-service portal for 24/7 status updates, with result records secured by the platform's tamper-evident HMAC-SHA256 audit logging. The platform alleviates the need to visit physically for inquiry by providing immediate answers through the chatbot, and other offline services.

Ilocos Sur Polytechnic State College. The institution gains a deployable reference implementation that operationalizes local capstone research into active administrative tools. This demonstrates how modern technical frameworks can solve specific operational failures within a state college environment, establishing an empirical blueprint that directly reconciles ambitious macroeconomic digitization mandates with the acute resource limitations of regional public institutions.

The Students. Incoming students validate that the admission exam process effectively reflects their true performance, ensuring that subsequent course placement is accurate and satisfying. This persistent profile remains securely accessible for personal verification and distributed cross-office institutional use without any record duplication.

The Researchers. The team develops advanced technical competencies in AI-assisted development, security architecture, and offline synchronization within a full project lifecycle. Methodologically, the study builds expertise in descriptive-developmental inquiry and empirical evaluation using the TAM framework.

Future Researchers. This study provides a technical blueprint for web-based, role-governed, unified, and offline-resilient platforms in constrained institutions. It establishes a framework for human-directed, AI-augmented software development, showing how small teams leverage generative models under strict public mandates. Additionally, it offers a replicable, and extensible model for transitioning machine learning from post-hoc analysis into live decision-support, enabling longitudinal verification of applicant profiling outcomes and other applied studies.

Chapter 2

METHODOLOGY

This chapter presents the procedures for data collection and analysis, covering the research design, software model, project plan, project assignments, population and locale, research instruments, and data analysis.

Research Design

This study employs a descriptive-developmental research design (Frey, 2022) to sequentially diagnose existing admission workflows at ISPSC Tagudin Campus and construct an evidence-based platform that resolves them. In the descriptive phase, data gathered through semi-structured administrator questionnaires and formally documented field observations undergo thematic analysis, structuring preliminary operational pain points into validated emergent themes. The developmental phase subsequently converts these finalized themes into the functional scope of NEXIAM, ensuring every integrated system requirement traces directly to a confirmed administrative failure

The developmental component encompasses the iterative design, construction, and validation of NEXIAM, addressing Specific Objectives 2, 3, and 4. System requirements extracted from the descriptive findings are organized into a functional requirements table and a system features table constituting the output of Specific Objective 2. The system is then constructed through the RAD workflow described in the following section and evaluated through simulated user acceptance testing with purposively selected evaluators, using the TAM questionnaire as the primary evaluation instrument.

Software Model

In developing institutional web systems, the Rapid Application Development (RAD) model (Martin, 1991; Pressman & Maxim, 2020) has proven effective for ensuring functional suitability and security, particularly in recent Philippine state university deployments (Olipas, 2020; Melendres & Aranda, 2024). This study advances traditional RAD methodology by systematically integrating AI-augmented development, aligning with recent Department of Science and Technology (2025) initiatives that equip State Universities and Colleges with generative AI to accelerate prototyping. This reflects what a 2026 Google whitepaper terms the transition from ad-hoc prompting to agentic engineering (Osmani et al., 2026). Under this paradigm, AI agents reliably generate the majority of routine implementation work, while developers act as system orchestrators focused entirely on architectural intent, specification quality, and complex edge cases. Crucially, to ensure generation velocity does not compromise system integrity, every AI-produced artifact is strictly governed by the workflow illustrated in Figure 2. This continuous feedback loop ensures human judgment controls the codebase prior to integration.

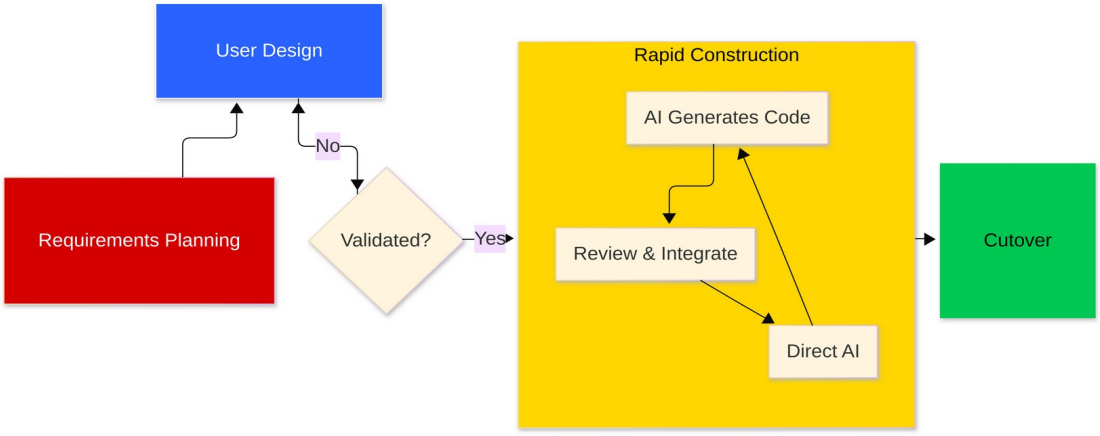


Figure 2: AI-Augmented Rapid Application Development (RAD) Model

Requirements Planning. This phase establishes an empirical baseline of institutional workflow gaps. It systematically gathers verified operational evidence through questionnaires and field observation to produce the formal technical specifications and the initial database schema. These validated operational failures become the highly specific local context used to ground all subsequent AI-assisted planning, preventing the generation of generic software solutions.

User Design. This phase translates the field-verified requirements into deterministic architectural blueprints. It formulates the relational Role-Based Access Control database schema, establishes the authorization matrix, architectural diagrams such as data flow, entity-relationship, and business requirements diagrams, and lastly, generates interface mockups. By finalizing these blueprints before coding begins, the team ensures the AI generates database structures that inherently comply with the Data Privacy Act (Republic Act No. 10173), structurally preventing the model from outputting non-compliant architectures.

Rapid Construction. This phase forms the technical development core, where the team systematically researches and evaluates potential application frameworks to accommodate complex system requirements. The team directs AI models to construct the heavy processing layers, such as the logic for computer-vision OMR scoring and the orchestration for the Retrieval-Augmented Generation chatbot. The developers conduct rigorous peer code reviews on every AI-generated module to ensure code quality. This AI-augmented workflow aligns with productivity gains documented in recent software engineering studies (Peng et al., 2023).

Cutover. This final phase transitions the development cycle into empirical

evaluation. It deploys the completed NEXIAM artifact into a secure cloud-hosted environment to fully demonstrate its operational capacity. Following deployment, the phase conducts simulated user testing with purposive selected panels and administers the paired Technology Acceptance Model evaluation protocol to measure system acceptance through rigorous statistical validation.

Project Plan

The project timeline spans four months, from June to September 2026, organized by the four RAD phases and presented in Figure 3 as a Gantt chart.

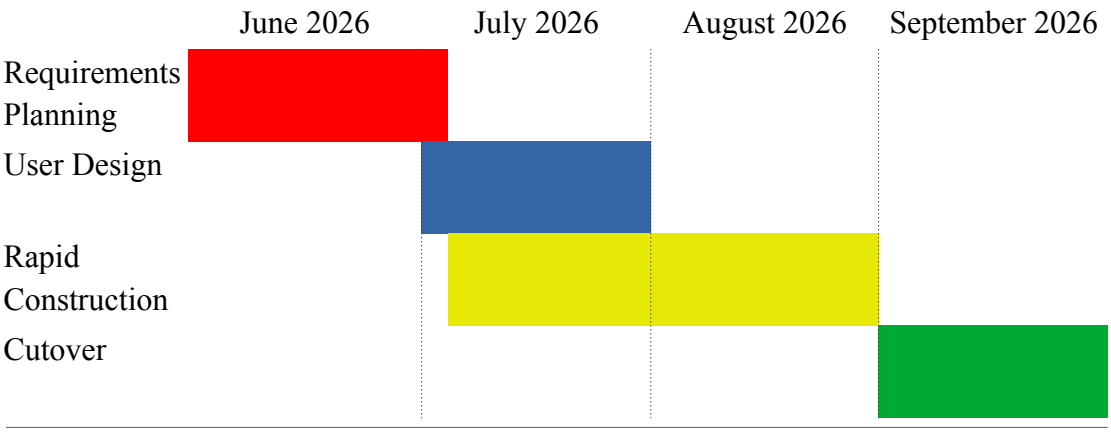


Figure 3: Project Gantt Chart

The Requirements Planning phase occupies June to early July 2026, covering administrator questionnaires, workflow observations, thematic analysis, formal specification drafting, and initial database schema. Findings from this phase are confirmed directly with the Guidance and Registrar Offices before the team proceeds to User Design, ensuring every documented pain point is validated by the office that actually experiences it rather than assumed from observation alone. User Design runs all throughout July 2026, producing the role-based schema, authorization matrix,

application service architecture, architectural diagrams, and interface mockups for all seven roles. This phase exists to validate feature needs before construction begins, rather than to accumulate more features.

Rapid Construction spans early July to August 2026 as the build-and-test core, structured as an iterative back-and-forth between design refinement and incremental construction rather than a single linear build, prioritizing core functionality over feature breadth wherever development time is constrained. Coding work is scheduled around the team's ongoing academic course load, including weekend sessions during heavier class weeks, and the extent of AI-assisted code generation is bounded by the team's own funding for model access. Cutover occupies September 2026, encompassing cloud-hosted deployment, TAM evaluation, and manuscript finalization and documentation.

This timeline deliberately allocates its largest continuous block to Rapid Construction. Because the team utilizes the agentic engineering methodology, the traditional timeline bottleneck of manual syntax generation is removed from the schedule. This specific logistical shift is what allows a compact three-person academic team to successfully execute heavy enterprise layers within a strict four-month operational window. By recovering hours that would otherwise be spent writing boilerplate code, the researchers reallocate their scheduling capacity directly into complex algorithmic tuning, such as building the integration layer for the K-Means aptitude model and future ML models, refining OMR for accuracy targets, and iterating on RAG Chatbot reliability and security. Ultimately, this approach secures vital integration testing time for the complex subsystems without compressing the

Cutover evaluation, thereby supporting stability for the following stages.

Project Assignments

Five project roles are distributed across the three team members, as detailed in Table 1. This cross-functional distribution is intentionally structured to maximize the operational capacity of a compact academic team. By establishing dual roles and overlapping development duties, the researchers ensure continuous project momentum, facilitate rigorous peer code review during complex technical integrations, and maintain strict adherence to the defined timeline. This formal matrix isolates system construction from objective testing to prevent internal evaluation bias.

Table 1: Project Roles and Responsibilities

Role	Name	Function
Project Manager	David Datu N. Sarmiento	Coordinates project timelines, stakeholder communication, and overall research execution across all RAD phases.
System Analyst and Designer	David Datu N. Sarmiento, Jaypee G. Pagaduan	Designs the system architecture, database schema, RBAC security models, and converts field requirements into technical specifications.
Lead Developer and Programmer	Jaypee G. Pagaduan	Implements backend services, frontend components, and specialized integrations (OMR, AI Features, Decision-support).
Quality Assurance Tester	Christine M. Lopez	Manages the evaluation protocol, administers the TAM instrument, and tracks system defects during the Cutover testing phase.
Technical Writer and Documenter	David Datu N. Sarmiento, Jaypee G. Pagaduan, Christine M. Lopez	Collaboratively drafts, edits, and finalizes the research manuscript, evaluation results, and formal system documentation.

Population and Locale of the Study

The study is conducted at the Ilocos Sur Polytechnic State College (ISPSC) Tagudin Campus, targeting the administrative offices and academic personnel managing the admission cycle. Because the September 2026 evaluation occurs outside the active admission period, actual applicants are unavailable, necessitating a purposive sampling design that carefully considers the venue constraints and the time allotment of the participants. The sample consists of 47 purposively selected evaluators chosen for their direct alignment with the institutional workflow. This includes Guidance Office staff (n=2) and Registrar Office personnel (n=3) who manage core administrative functions, Program Heads (n=11) responsible for course triage and local program qualification standards, and first-year students (n=30) evaluating the applicant portal. Utilizing first-year students brings significant relevance to the study, as they have just recently undergone the actual institutional admission process and possess fresh, direct experience with its local requirements. Similarly, the selected Guidance and Registrar personnel represent the complete available staff complement of their respective offices at the Tagudin Campus; because they intimately know the existing admission bottlenecks, their detailed walkthrough of the system workflow provides an authentic, high-fidelity institutional evaluation. Finally, the evaluation includes one (n=1) IT expert who independently reviews the underlying system architecture, role-based security matrices, and data privacy compliance protocols to ensure institutional data integrity.

Table 2: Distribution of Respondents

Respondents	n
First Year Students	30
Registrar Office Personnel	3
Guidance Office Personnel	2
Program Head	11
IT Expert	1
Total	47

Research Instruments

This study utilizes an extended Technology Acceptance Model (TAM) questionnaire as its primary evaluation instrument. Grounded in Davis (1989), the tool measures perceived usefulness and perceived ease of use. It integrates attitude toward use and behavioral intention to use, following Venkatesh and Davis (2000) and King and He (2006). This extended TAM framework has been empirically validated for assessing digital service transitions specifically within Philippine higher education institutions (Ampuan & Deleña, 2024; Quiban, 2025). Evaluating these constructs pinpoints potential user resistance points regarding the adoption of novel system features like automated scheduling and OMR scoring.

To ensure construct validity, the 16-item instrument employs role-adapted phrasing across four respondent groups detailed in Tables 3 through 6: Registrar Personnel, Guidance Personnel, Applicants, and Program Heads. Indicators are tailored to the operational realities of each group while preserving underlying

psychometric properties. However, TAM's baseline constructs are in nature fixed, thus the practice of item adaptation preserves content validity across these roles. Safie et al. (2025) validated this integrated TAM approach, tailoring items to individual practitioners for usability and utility versus organizational roles for readiness and governance. Items are measured on a seven-point Likert scale (1 to 7), with four items per construct. This provides greater granular variance than five-point alternatives, capturing subtle shifts in perception between manual and digital workflows.

Methodologically, a within-subject, parallel object design requires evaluators to complete the instrument twice: once, as a pre-test assessing the baseline manual process and second, a post-test after system use. The respondent recruitment targets individuals with direct, active recall of the admission process, formally tracing any reassigned staff to ensure accurate baseline data. This paired structure neutralizes between-group variability, yielding the empirical comparison required to test the null hypotheses in Specific Objective 4. Contrasting baseline paper inefficiencies against post-test data directly quantifies system success in resolving bottlenecks.

In addition to the TAM instrument, the single IT expert evaluates NEXIAM through a structured technical checklist covering four areas: system architecture soundness, role-based access control implementation, application-layer data privacy compliance aligned with Republic Act No. 10173, and overall maintainability. This technical review is reported qualitatively alongside the TAM results and serves as supplementary validation of the system's design integrity. Because n=1, no inferential statistics are applied to the IT expert's responses.

Table 3: TAM Indicators on Perceived Usefulness (PU) for Registrar Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	The current process allows me to process applicant intake and registration efficiently.	NEXIAM allows me to process applicant intake and registration efficiently.
2	The current process helps me manage admission-related records faster.	NEXIAM helps me manage admission-related records faster.
3	The current process gives me sufficient visibility into an applicant's status when needed.	NEXIAM gives me sufficient visibility into an applicant's status when needed.
4	The current process reduces my overall workload when coordinating with the guidance personnel.	NEXIAM reduces my overall workload when coordinating with the guidance personnel.

Table 4: TAM Indicators on Perceived Usefulness (PU) for Guidance Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	I have the tools to profile applicants based on their exam performance.	NEXIAM's decision-support triage helps me evaluate applicant readiness for placement.
2	The current process of reviewing exam scores is adequate for my counseling sessions.	NEXIAM's triage output provides useful profiles for counseling sessions.
3	The current process gives me sufficient visibility into an applicant's status when needed.	NEXIAM gives me sufficient visibility into an applicant's status when needed.
4	The current process reduces my overall workload when coordinating with the registrar and program head personnel.	NEXIAM reduces my overall workload when coordinating with registrar and program head personnel.

Table 5: TAM Indicators on Perceived Usefulness (PU) for Applicant

Item	Pre-Test Indicator	Post-Test Indicator
1	The current way of communicating updates and results is fast and helpful.	The NEXIAM portal and chatbot are fast and helpful for communicating updates and results.
2	The current way of submitting my application is convenient.	The NEXIAM portal makes submitting my application convenient.
3	The current process keeps me adequately informed about my application status.	NEXIAM keeps me adequately informed about my application status.
4	The current process makes it easy for me to verify that my exam results are accurate.	NEXIAM's results portal makes it easy for me to verify that my exam results are accurate.

Table 6: TAM Indicators on Perceived Usefulness (PU) for Program Head

Item	Pre-Test Indicator	Post-Test Indicator
1	The current informal method of communicating quota updates effectively supports student placement.	The real-time quota visibility in NEXIAM effectively supports student placement.
2	My current level of visibility regarding incoming applicants helps me make timely program decisions.	The pre-enrollment slate in NEXIAM helps me make timely program decisions.
3	The current method of coordinating with guidance personnel is productive.	Coordinating with guidance personnel through NEXIAM is productive.
4	The current method gives me an accurate picture of my department's enrollment capacity.	NEXIAM gives me an accurate picture of my department's enrollment capacity.

Table 7: TAM Indicators on Perceived Ease of Use (PEOU) for Registrar Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	It is easy to navigate the current intake forms and paper filing.	It is easy to navigate the digital applicant intake and records in NEXIAM.
2	Learning how to carry out the current intake and filing procedures took little effort.	Learning how to use NEXIAM took little effort.
3	It is easy for me to correct mistakes when using the current process.	It is easy for me to correct mistakes in applicant records using NEXIAM.
4	Interacting with the current process is generally clear and understandable.	Interacting with NEXIAM is generally clear and understandable.

Table 8: TAM Indicators on Perceived Ease of Use (PEOU) for Guidance Personnel.

Item	Pre-Test Indicator	Post-Test Indicator
1	It is easy to navigate the current scheduling, stencil scoring, and profiling.	It is easy to navigate the AI-assisted scheduling, automated OMR scoring, and triage in NEXIAM.
2	Learning how to carry out the current scheduling and scoring procedures took little effort.	Learning how to use NEXIAM took little effort.
3	It is easy to correlate exam scores with program requirements using the current process.	It is easy to read and interpret the triage profiles in NEXIAM.
4	Interacting with the current process is generally clear and understandable.	Interacting with NEXIAM is generally clear and understandable.

Table 9: TAM Indicators on Perceived Ease of Use (PEOU) for Applicant

Item	Pre-Test Indicator	Post-Test Indicator
1	Completing the admission steps through the current physical process is easy to understand.	Completing the admission steps through the NEXIAM portal is easy to understand.
2	It is easy for me to find the information I need during the current admission process.	It is easy for me to find the information I need on the NEXIAM portal.
3	Becoming skillful at the current admission process requires little effort on my part.	Becoming skillful at using NEXIAM requires little effort on my part.
4	The steps required by the current process are clear at every stage.	The steps required by NEXIAM are clear at every stage.

Table 10: TAM Indicators for Perceived Ease of Use (PEOU) on Program Head

Item	Pre-Test Indicator	Post-Test Indicator
1	It is easy to track and communicate enrollment quota changes to counselors.	It is easy to update and monitor enrollment quotas through NEXIAM.
2	Anticipating incoming student routing through the current method is simple.	Reviewing the pre-enrollment slate through the NEXIAM dashboard is simple.
3	The current method gives me adequate insight into applicant aptitude before they enroll	NEXIAM gives me adequate insight into applicant aptitude before they enroll
4	The current method is flexible enough to interact with the way I want.	NEXIAM is flexible enough to interact with the way I want.

Table 11: TAM Indicators on Attitude Toward Use (ATU) for Registrar Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	I have a favorable attitude toward the current admission workflow.	I have a favorable attitude toward using the NEXIAM platform.
2	Using the current process is a good idea for managing applicant intake and records.	Using NEXIAM is a good idea for managing applicant intake and records.
3	I find the current filing and record-keeping to be a pleasant way of getting work done.	I find NEXIAM to be a pleasant way of getting work done.
4	Overall, I am satisfied with the current process.	Overall, I am satisfied with NEXIAM.

Table 12: TAM Indicators on Attitude Toward Use (ATU) for Guidance Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	I have a favorable attitude toward the current admission workflow.	I have a favorable attitude toward using the NEXIAM platform.
2	Using the current process to schedule exams and profile applicants is a good approach.	Using NEXIAM to schedule exams and profile applicants is a good approach.
3	I find the current scheduling, scoring, and profiling to be a pleasant part of my work.	I find NEXIAM's scheduling, scoring, and triage dashboard to be a pleasant part of my work.
4	Overall, I am satisfied with the current process.	Overall, I am satisfied with NEXIAM.

Table 13: TAM Indicators for Attitude Toward Use (ATU) on Applicant

Item	Pre-Test Indicator	Post-Test Indicator
1	I feel positive about my experience with the current in-person admission process.	I feel positive about my experience using the automated NEXIAM process.
2	Using the current process to apply is a good way to handle my application.	Using NEXIAM to apply is a good way to handle my application.
3	I find the current admission process to be a pleasant experience overall.	I find NEXIAM to be a pleasant experience overall.
4	Overall, I am satisfied with the current admission process.	Overall, I am satisfied with NEXIAM.

Table 14: TAM Indicators on Attitude Toward Use (ATU) for Program Head

Item	Pre-Test Indicator	Post-Test Indicator
1	I have a favorable attitude toward using informal channels (e.g., texts, calls) for quota updates.	I have a favorable attitude toward using the NEXIAM dashboard for quota updates.
2	Relying on my current level of applicant visibility is a good approach for my department.	Relying on the NEXIAM pre-enrollment slate is a good approach for my department.
3	I find the current method of receiving updates to be a pleasant part of my work.	I find NEXIAM to be a pleasant part of my work.
4	Overall, I am satisfied with the current method of managing quota communication.	Overall, I am satisfied with NEXIAM for managing quota communication.

Table 15: Table 6.1. TAM Indicators on Behavioral Intention to Use (BIU) for Registrar Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	I would prefer to keep using the current process for future admission cycles.	I intend to use NEXIAM for future admission intake and record management.
2	I would recommend continuing the current process to colleagues in other offices.	I would recommend NEXIAM to colleagues in other offices.
3	I plan to keep relying on the current filing and records for upcoming admissions.	I plan to rely on NEXIAM for upcoming admission intake and records.
4	Given the choice, I would continue using the current process.	Given the choice, I would continue using NEXIAM.

Table 16: TAM Indicators on Behavioral Intention to Use (BIU) for Guidance Personnel

Item	Pre-Test Indicator	Post-Test Indicator
1	I would prefer to continue the current scheduling, scoring, and profiling for future admission cycles.	I intend to use NEXIAM's scheduling, scoring, and triage for future admission cycles.
2	I would recommend continuing the current process to colleagues in other offices.	I would recommend NEXIAM to colleagues in other offices.
3	I plan to keep relying on the current scheduling, scoring, and profiling for upcoming admissions.	I plan to rely on NEXIAM for upcoming scheduling, scoring, and admissions counseling.
4	Given the choice, I would continue using the current process.	Given the choice, I would continue using NEXIAM.

Table 17: TAM Indicators for Behavioral Intention to Use (BIU) of Applicant

Item	Pre-Test Indicator	Post-Test Indicator
1	I would recommend the current manual process to future incoming freshmen.	I would highly recommend NEXIAM to future incoming freshmen.
2	I would choose to go through the current process again if I had to reapply.	I would choose to use NEXIAM again if I had to reapply.
3	I intend to continue following the current process for any future transactions with the institution.	I intend to continue using NEXIAM for any future transactions with the institution.
4	I plan to rely on the current process to stay updated on my application.	I plan to rely on NEXIAM to stay updated on my application.

Table 18: TAM Indicators for Behavioral Intention to Use (BIU) of Program Head

Item	Pre-Test Indicator	Post-Test Indicator
1	I intend to continue using informal communication for department quota management.	I intend to use the NEXIAM dashboard for department quota management in future admission cycles.
2	I would recommend the current way of tracking incoming applicants to other Program Heads.	I would recommend NEXIAM's pre-enrollment slate to other Program Heads.
3	I plan to keep using the current method for future admission cycles.	I plan to keep using NEXIAM for future admission cycles.

Item	Pre-Test Indicator	Post-Test Indicator
4	Given the choice, I would continue relying on the current method.	Given the choice, I would continue relying on NEXIAM.

Data Analysis

For Specific Objective 1, qualitative thematic analysis is applied to questionnaire responses from Guidance and Registrar staff. Responses are transcribed, coded, and analyzed to structure the operational themes observed during field inquiry: scoring bottlenecks, inter-office coordination failures, data integrity vulnerabilities, and notification deficiencies. Researcher field observation notes supply the procedural context that questionnaire data alone cannot fully capture. The resulting thematic structure constitutes the empirical foundation from which every NEXIAM feature is derived: no requirement enters development without a documented operational failure.

For Specific Objective 2, design validation proceeds iteratively during the construction cycle. Completed modules are demonstrated to Guidance and Registrar staff, and any misalignment between the system's design and field-confirmed needs is logged as a revision item, connecting every design change to a specific field finding. The functional requirements and system features tables simultaneously constitute Chapter 3's system design section and the reference structure against which Cutover testing scenarios are built.

For Specific Objective 3, each respondent's four items per construct are averaged into a single per-person construct score; these per-person scores are then

averaged across all evaluator post-test responses to produce the pooled mean for each TAM construct. Means above the midpoint of the seven-point scale (>4.0) indicate positive acceptance; standard deviations per construct are reported alongside the means to describe response consistency. Pooled means are reported alongside a breakdown of per-role descriptive means (Registrar, Guidance, Applicant, Program Head) for each construct. Results are interpreted against established TAM benchmarks from the original instrument and its subsequent validation literature.

The IT expert's technical review findings are reported qualitatively as a standalone subsection within the evaluation results, covering architecture, security, data privacy, and maintainability. These findings do not enter the pooled TAM means or the paired pre-post comparison; they serve as supplementary expert validation that the system meets institutional technical standards.

For Specific Objective 4, a paired samples t-test compares pre-test and post-test scores for perceived usefulness and perceived ease of use at a significance threshold of $\alpha = 0.05$. However, because the sample size is relatively small ($n = 46$), that is also excluding the IT expert, a Shapiro-Wilk test evaluates the normality distribution of the paired differences. If the normality assumption is violated, the Wilcoxon signed-rank test is deployed as the robust non-parametric alternative to ensure statistical validity. Effect size is reported as Cohen's d when the paired t-test is used, or as the matched-pairs rank-biserial correlation when the Wilcoxon signed-rank test is used instead.

Glossary

Admission Testing System. The end-to-end digital platform NEXIAM provides, covering applicant registration, examination administration, automated scoring, decision-support triage, and result delivery that is governed by role-based access control.

Applicant Portal. The NEXIAM web interface through which prospective students register, submit requirements, track application status, and view examination results without requiring physical presence on campus.

Aptitude Pattern. A descriptive clustering of examination subtest scores that reveals natural groupings in applicant performance. In NEXIAM, aptitude patterns are surfaced by the K-Means module as a visualization aid for counselor-led placement decisions. The algorithm surfaces the pattern; the counselor makes the placement.

AI-Augmented Rapid Application Development (AI-RAD). The software development methodology adopted for constructing NEXIAM. It integrates AI code generation into the standard RAD phases while enforcing a human Review and Integrate gate on every AI-produced artifact.

Audit Trail. A chronologically chained record of system events whose integrity is enforced through HMAC-SHA256 cryptographic hashing. Each entry includes the hash of the previous entry, making any post-hoc tampering detectable.

Data Privacy Act (RA 10173). The Philippine law mandating protection of personal data in both government and private sector processing. NEXIAM is designed to align with RA 10173 at the application layer through RBAC-enforced data isolation

and HMAC-SHA256 audit logging. Formal National Privacy Commission certification is outside scope.

Free Higher Education Act (RA 10931). The 2017 Philippine law providing free tuition in all state universities and colleges. It significantly increased admission volumes at ISPSC, creating the operational pressure that motivates NEXIAM's digitization of the admission testing pipeline.

HMAC-SHA256. A cryptographic function combining a secret key with message data to produce a tamper-evident hash. In NEXIAM, every audit log entry carries an HMAC-SHA256 signature chained to the previous entry's hash, ensuring any unauthorized modification is structurally detectable.

Human-Gated Approval. A design constraint in NEXIAM where no automated recommendation — such as an AI-proposed exam schedule — takes effect until a human actor explicitly reviews and approves it. The system proposes; the authorized officer disposes.

Input-Process-Output (IPO) Model. The conceptual framework organizing NEXIAM's research design. Input captures institutional data entering the pipeline. Process covers RAD development and the paired TAM evaluation. Output consists of the NEXIAM platform and the empirical results from the pre-test/post-test comparison

K-Means Clustering. An unsupervised machine learning algorithm that partitions applicant subtest scores into descriptive aptitude groups. In NEXIAM, K-Means surfaces patterns for counselor review. It is a visualization and decision-

support tool — it does not make or recommend admission decisions.

Likert Scale. The 7-point response format (1 = Strongly Disagree to 7 = Strongly Agree) used in NEXIAM's TAM questionnaire. The 7-point range provides greater variance than five-point alternatives for detecting the pre-to-post perception shift that Specific Objective 4 measures.

NEXIAM. The proposed system. The acronym stands for Next-Generation Examinations and (X) Cross-Role Intelligent Admissions Management. NEXIAM is a web-based, role-governed college admission testing platform integrating RBAC, computer-vision OMR, K-Means aptitude clustering, a RAG chatbot, AI-assisted scheduling, and HMAC-SHA256 audit logging.

Offline Resilience. The proctoring interface's ability to maintain examination session state during connectivity loss through service-worker caching, with automatic synchronization to the cloud database when the connection restores. This applies only to the proctoring module; all other system functions require cloud connectivity.

Optical Mark Recognition (OMR). The computer-vision pipeline in NEXIAM that automates scoring of physical answer sheets. It applies perspective correction to compensate for scanner skew, then performs contour detection to read marked bubbles — replacing a manual process that currently takes five days to two weeks.

Paired T-Test. The primary parametric statistical test used in Specific Objective 4 to compare pre-test and post-test TAM scores. It is applied only if the Shapiro-Wilk test confirms normality of the paired differences. If normality is violated, the Wilcoxon signed-rank test is used instead.

Post-Test. The second administration of the TAM questionnaire, conducted after evaluators have used NEXIAM in simulated operational scenarios. It captures user perception of the system to be compared against the Pre-Test baseline.

Pre-Test. The initial administration of the TAM questionnaire, capturing evaluator perceptions of the existing manual admission process before they interact with NEXIAM. It establishes the baseline against which Post-Test scores are compared.

Proctor. A duty overlay assignable to any existing staff role at administrative discretion — not a distinct eighth user role. The Proctor overlay grants access to the proctoring interface for attendance marking and session monitoring during examinations.

Purposive Sampling. The non-probability sampling method used to select NEXIAM's 47 evaluators. Participants are chosen for their direct operational knowledge: Registrar and Guidance staff who manage the current manual workflow, Program Heads responsible for course triage, first-year students with recent admission experience, and one IT expert evaluating backend architecture.

Rapid Application Development (RAD). The iterative software development methodology — structured around Requirements Planning, User Design, Rapid Construction, and Cutover — that governs NEXIAM's construction cycle. In this study, RAD is augmented with AI code generation under a mandatory human review gate.

Retrieval-Augmented Generation (RAG). A natural language processing

architecture that grounds an LLM's responses by first retrieving relevant chunks from a curated knowledge base. NEXIAM's chatbot uses RAG restricted to institutional policy documents — if no document matches the query above a confidence threshold, the chatbot responds that it does not have the information rather than fabricating.

Role-Based Access Control (RBAC). The security model governing all seven NEXIAM user roles: Applicant, Program Head, Guidance Counselor, Guidance Administrator, Registrar Staff, Registrar Administrator, and System IT Administrator. Each role receives only the permissions necessary for its designated functions. An eighth actor — Proctor — is a duty overlay assignable to any existing staff role.

Service Worker. A browser-side background script enabling the proctoring interface to cache examination session data locally during connectivity interruptions and synchronize to the cloud database upon reconnection.

Shapiro-Wilk Test. The normality test applied to paired TAM score differences before selecting the statistical test for Specific Objective 4. If the differences are normally distributed, the paired t-test is used. If not, the Wilcoxon signed-rank test is applied.

Technology Acceptance Model (TAM). The evaluation framework — grounded in Davis (1989) and extended by Venkatesh and Davis (2000) and King and He (2006) — measuring four constructs: Perceived Usefulness, Perceived Ease of Use, Attitude Toward Use, and Behavioral Intention to Use. NEXIAM's 16-item TAM instrument is administered as both a Pre-Test and Post-Test on a 7-point Likert scale, with items adapted to the operational realities of each respondent role.

Thematic Analysis. The qualitative method used in Specific Objective 1 to extract operational pain points from Guidance and Registrar staff interviews. Responses are coded and organized into themes that directly determine NEXIAM's functional requirements.

User Role. A defined set of RBAC permissions assigned to a category of system users. NEXIAM defines seven distinct roles — Applicant, Program Head, Guidance Counselor, Guidance Administrator, Registrar Staff, Registrar Administrator, and System IT Administrator — with Proctor as a duty overlay assignable to any staff role.

Wilcoxon Signed-Rank Test. The non-parametric alternative to the paired t-test for Specific Objective 4, applied when the Shapiro-Wilk test indicates that the paired differences are not normally distributed. Effect size is reported as the matched-pairs rank-biserial correlation.

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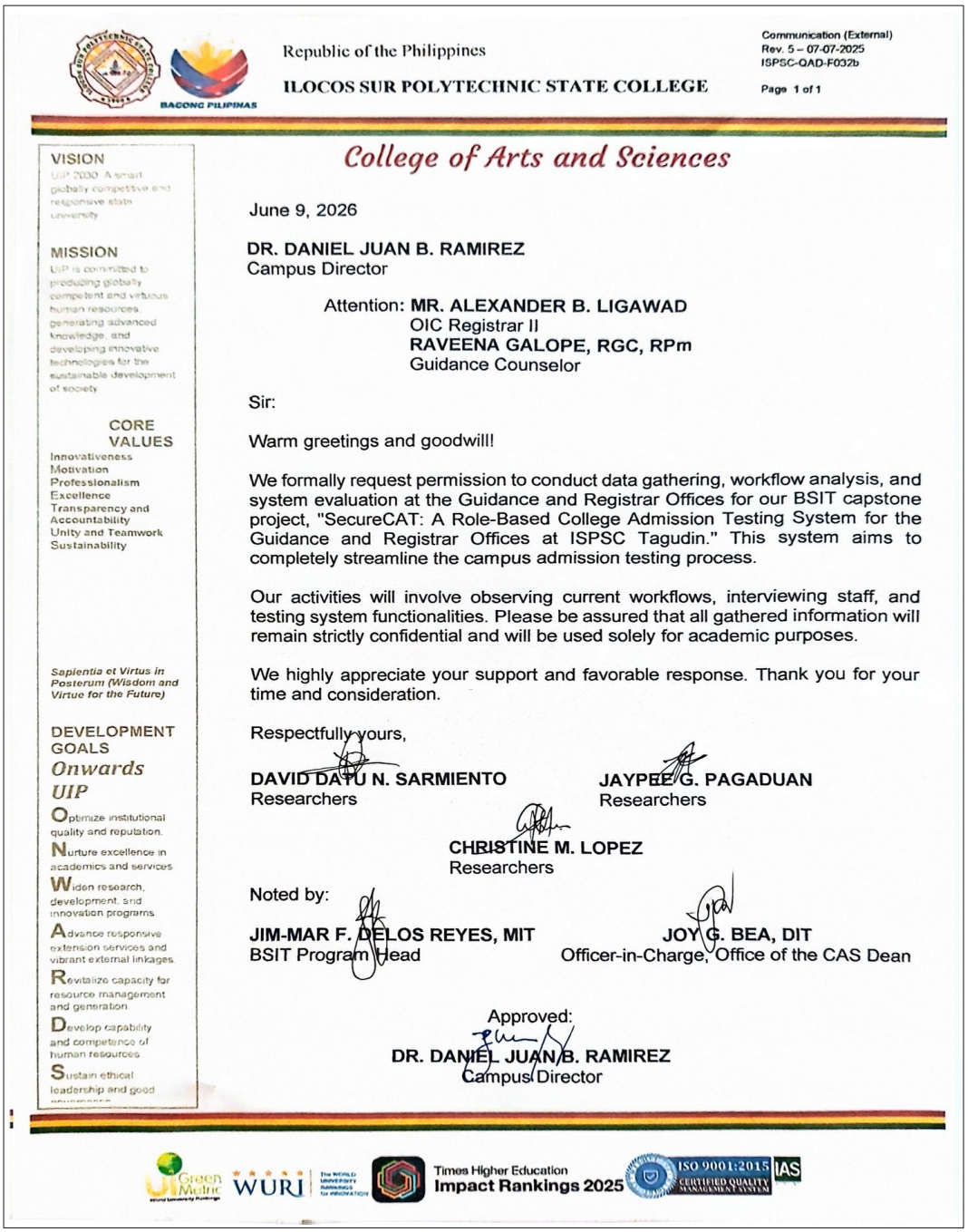
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APPENDICES

APPENDIX A

SCAN OF SIGNED LETTER OF CONDUCT

Given the revision of the study concerning its brand name as advised by its panelist, the name NEXIAM is put in place in replacement of SECURECAT.



Republic of the Philippines
ILOCOS SUR POLYTECHNIC STATE COLLEGE
BAGONG PILIPINAS

Communication (External)
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VISION
UIP 2030: A smart, globally competitive and responsive state university.

MISSION
UIP is committed to producing globally competent and virtuous human resources, generating advanced knowledge, and developing innovative technologies for the sustainable development of society.

CORE VALUES
Innovativeness
Motivation
Professionalism
Excellence
Transparency and Accountability
Unity and Teamwork
Sustainability

Sapientia et Virtus in Posterum (Wisdom and Virtue for the Future)

DEVELOPMENT GOALS
Onwards UIP
Optimize institutional quality and reputation.
Nurture excellence in academics and services.
Widen research, development, and innovation programs.
Advance responsive extension services and vibrant external linkages.
Revitalize capacity for resource management and generation.
Develop capability and competence of human resources.
Sustain ethical leadership and good

College of Arts and Sciences

June 9, 2026

DR. DANIEL JUAN B. RAMIREZ
Campus Director

Attention: **MR. ALEXANDER B. LIGAWAD**
OIC Registrar II
RAVEENA GALOPE, RGC, RPm
Guidance Counselor

Sir:

Warm greetings and goodwill!

We formally request permission to conduct data gathering, workflow analysis, and system evaluation at the Guidance and Registrar Offices for our BSIT capstone project, "SecureCAT: A Role-Based College Admission Testing System for the Guidance and Registrar Offices at ISPSC Tagudin." This system aims to completely streamline the campus admission testing process.

Our activities will involve observing current workflows, interviewing staff, and testing system functionalities. Please be assured that all gathered information will remain strictly confidential and will be used solely for academic purposes.

We highly appreciate your support and favorable response. Thank you for your time and consideration.

Respectfully yours,

DAVID DAYU N. SARMIENTO
Researchers

JAYPEE G. PAGADUAN
Researchers

CHRISTINE M. LOPEZ
Researchers

Noted by:

JIM-MAR F. DELOS REYES, MIT
BSIT Program Head

JOY G. BEA, DIT
Officer-in-Charge, Office of the CAS Dean

Approved:

DR. DANIEL JUAN B. RAMIREZ
Campus Director

UI GreenMetric WURI Times Higher Education Impact Rankings 2025 ISO 9001:2015 IAS

APPENDIX B

USE CASE DIAGRAM

The use case diagram maps NEXIAM's eight actors namely: Applicant, Proctor, Guidance Counselor, Program Head, System IT Administrator, Registrar Staff, Registrar Admin, and Guidance Admin against eighteen use cases covering intake, testing, decision-support triage, and audits. Access is strictly role-isolated where actors only access functions their offices perform, with the System IT Administrator managing overall function and accounts.

